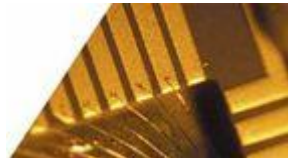


MEMS sensor packaging

Per-Erik Fägerman
ceo

Mandalon Technologies AB
Sweden



Talk outline

Company presentation

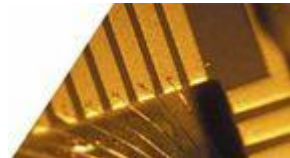
Presentation

"MEMS sensor packaging"

- The NORD-pie project
- R&D packaging approach
- Structures & packaging solutions
 - Ultrasonic ID transducer
 - Accelerometer
 - Gas sensor
 - Fish growth monitor

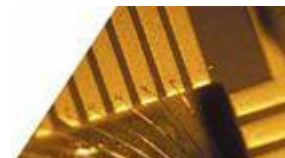
Conclusions

Questions

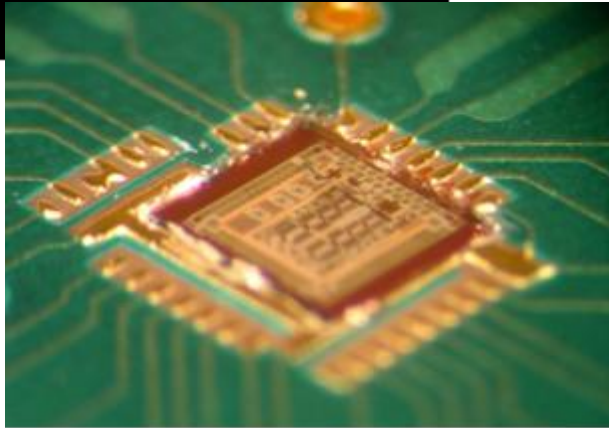
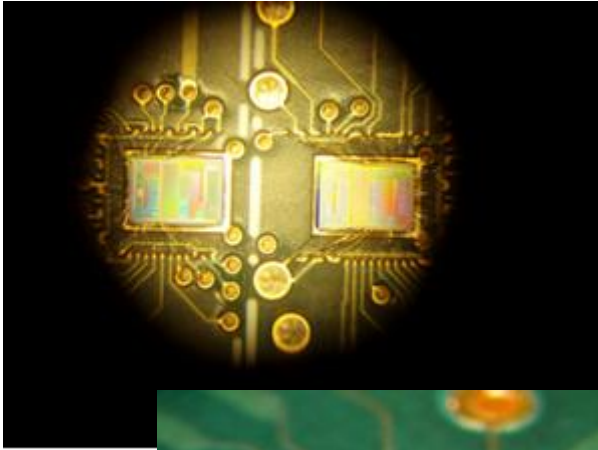


Mandalon Technologies AB

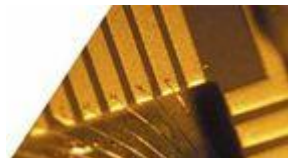
- Established 1999
- Background in university research, S-Sence
- Unique skills in chip packaging
- Prototypes and production series up to 5-10 thousands of units
- Chip Packaging Partner in European Research Projects



Competence for packaging

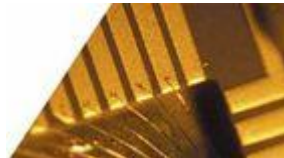


- More than 10 years in business
- Over 100 unique packaging solutions
- Several 1000 hours of bonding work



It is important for us...

- ...to build long lasting, developing relations with our customers
- ...to participate in international research projects to enhance our competence
- ...to perform professionally even though we are a small company, therefore the ISO 9001 certificate
- ...to show flexibility, creativity and quick response



Talk outline

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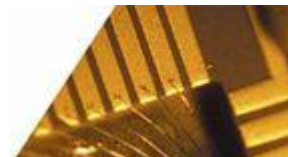
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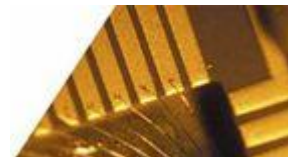
NORD-pie project

- Feasibility study of piezoMEMS devices
 - Design and prototyping
 - First industrial run of the SINTEF moveMEMS process

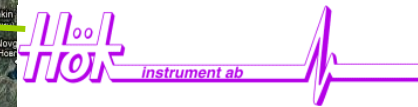
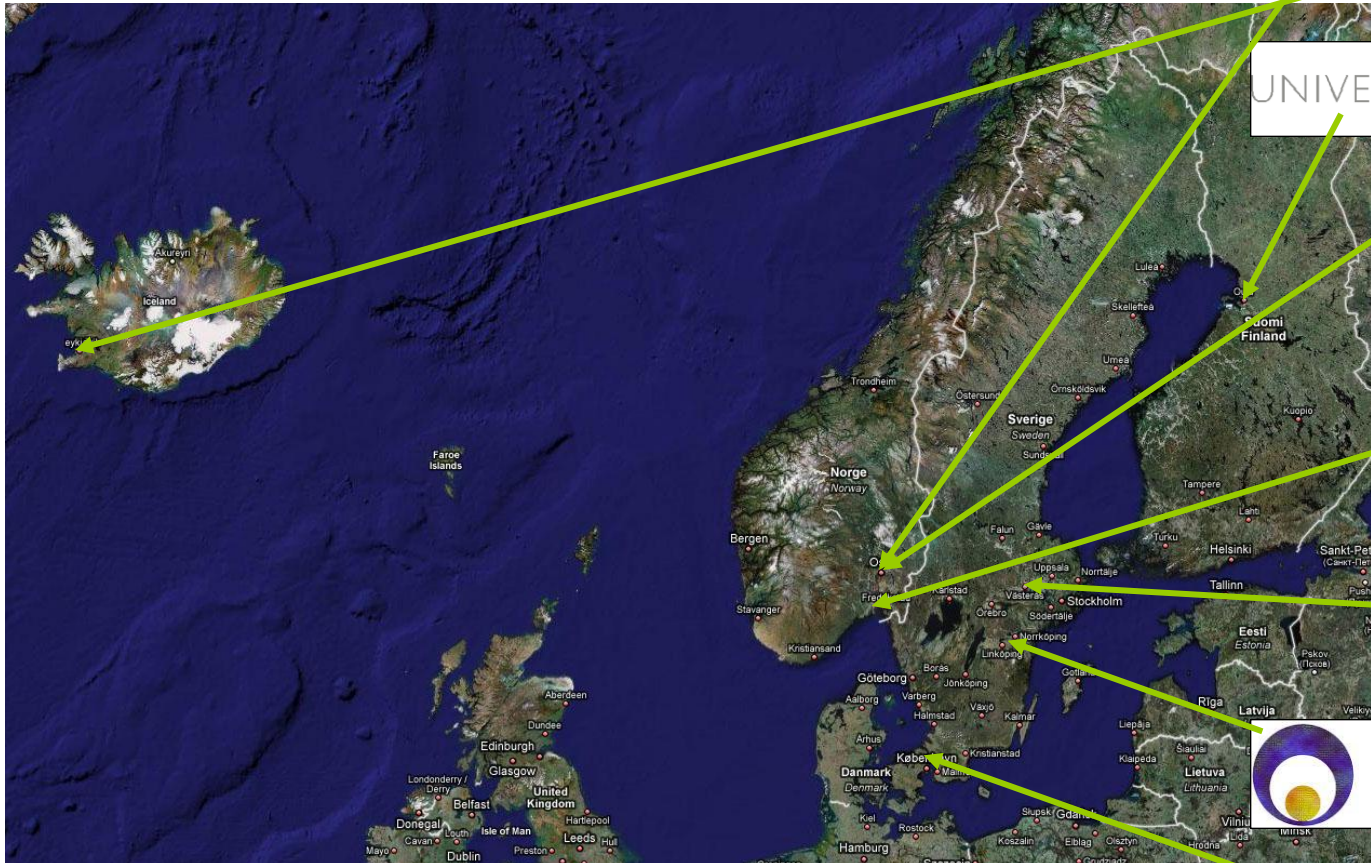
Support from Nordic Innovation Center, NiCe



IMAPS Sep 15 2008



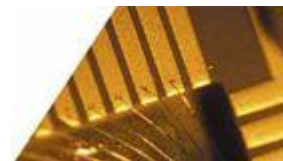
NORD-pie project



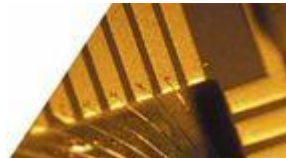
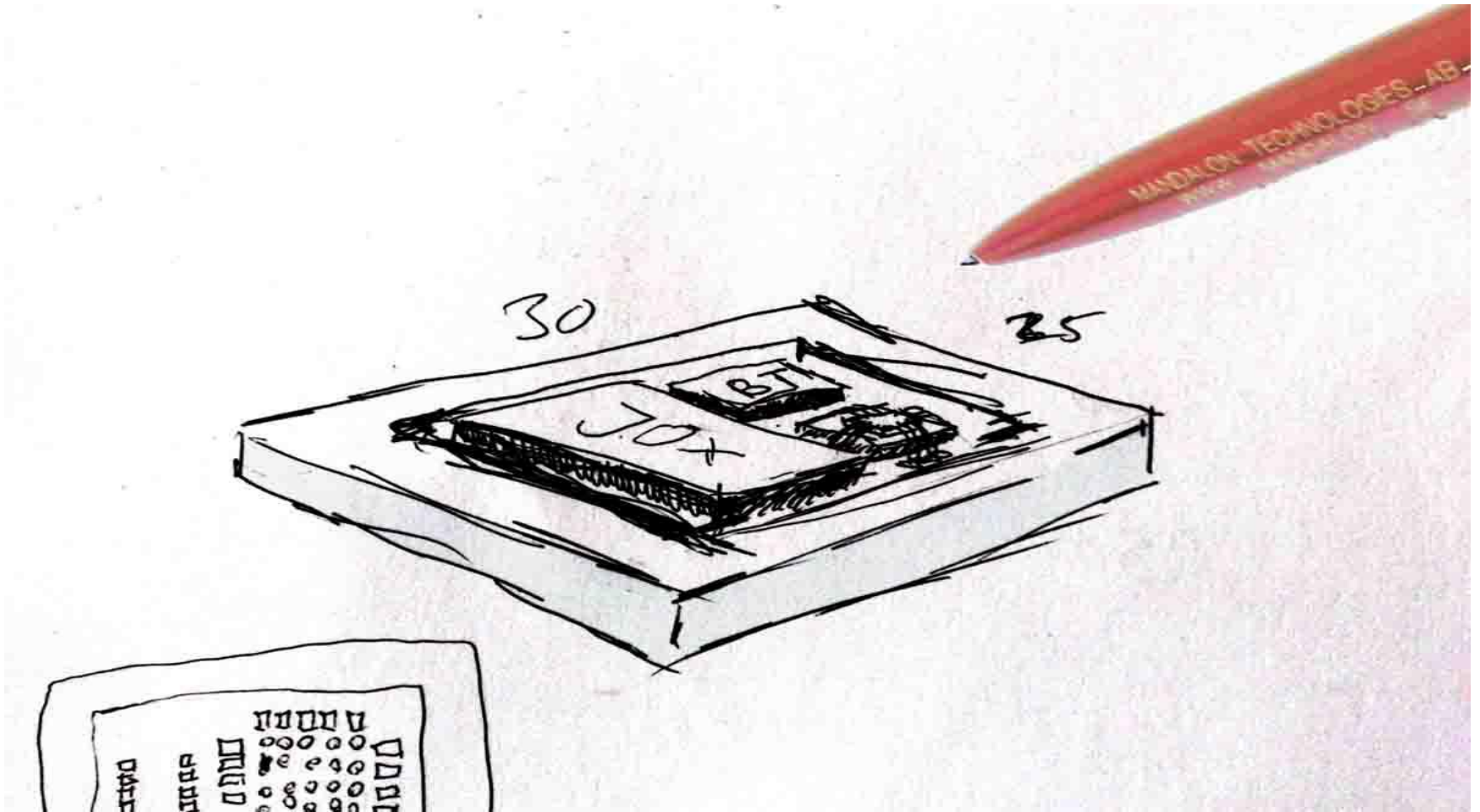
NORD-pie link is easily found on our home page



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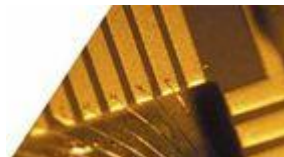


Packaging really starts before the packaging - with the design ideas



R&D packaging approach

- Design phase and system integration
- First draft version 1.0 for evaluation
- Feed back from users, tests etc
- Updated design in version 2.0



4 applications in the NORD-pie project

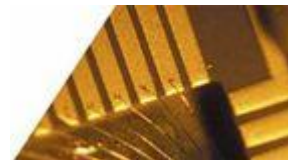
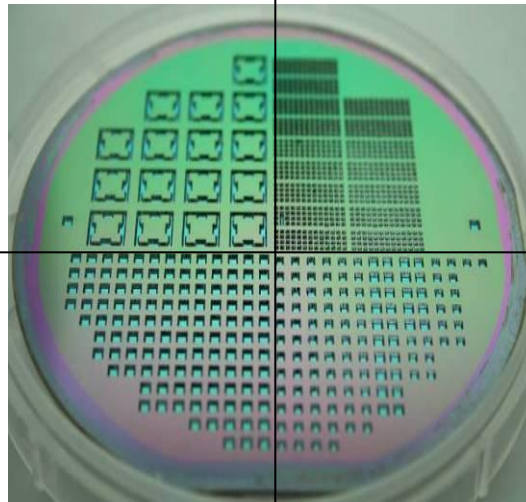
Thin film piezo chip packaging

Accelerometers
InSensor

Fish tags
Star-Oddi

Microphone
arrays
Sonitor

Acoustic gas sensors
Hök Instrument &
Infineon/SensoNor



4 different types of initial problems

Accelerometer

“Free” central mass,
very brittle
silicon bridges

Fish Tag

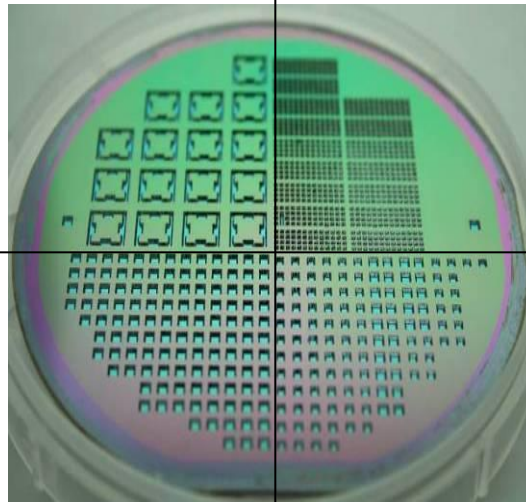
Many I/O's, water under
high pressure

Microphone arrays

Shielding off the
ultra sound from
back side to front

Acoustic gas sensors

Adjustable gas volume
within the package



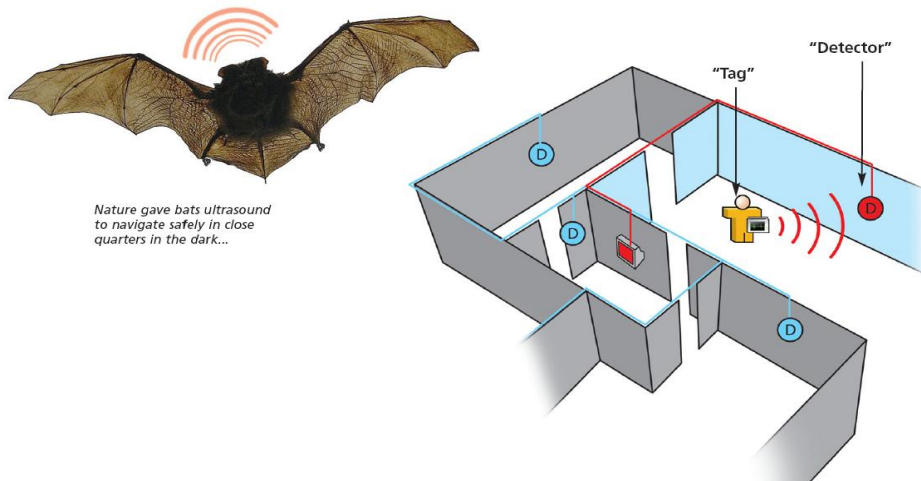
Mandalon
Technologies AB

IMAPS Sep 15 2008

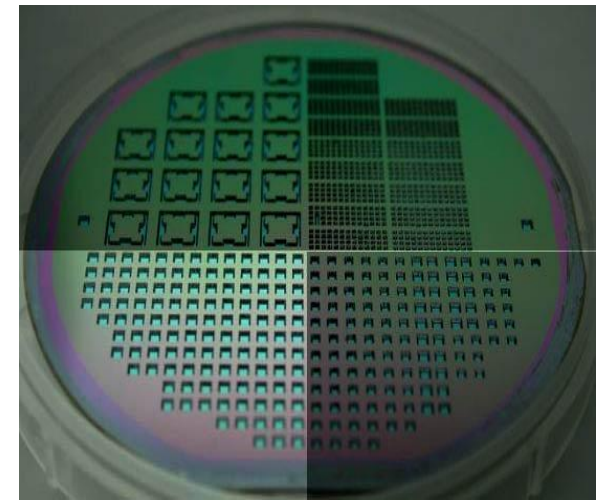


Ultrasonic microphone array prototype version 1.0

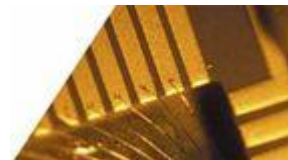
Objective: USID



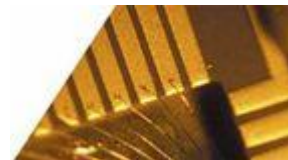
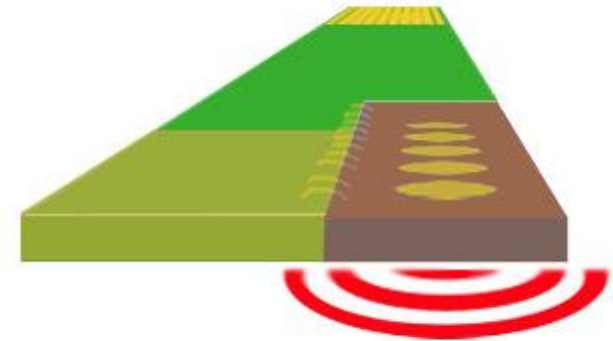
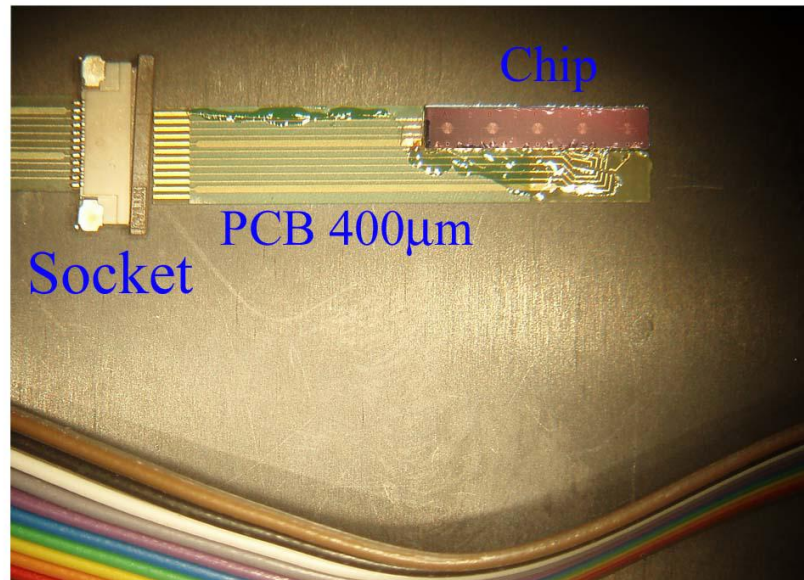
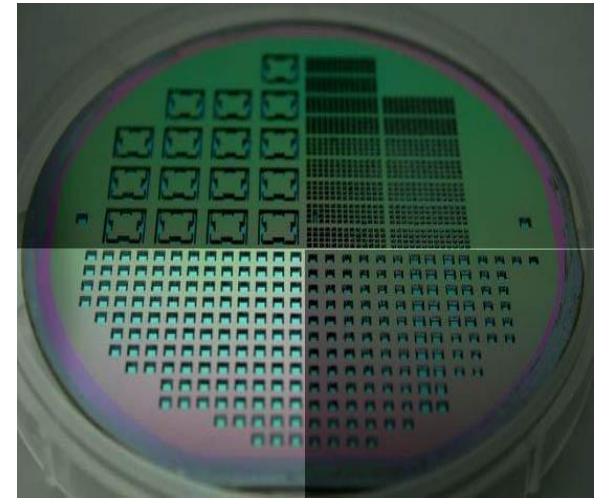
www.sonitor.com



To identify the
location of objects
within specific rooms



Ultrasonic microphone array prototype version 1.0



Ultrasonic microphone array

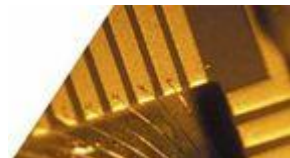
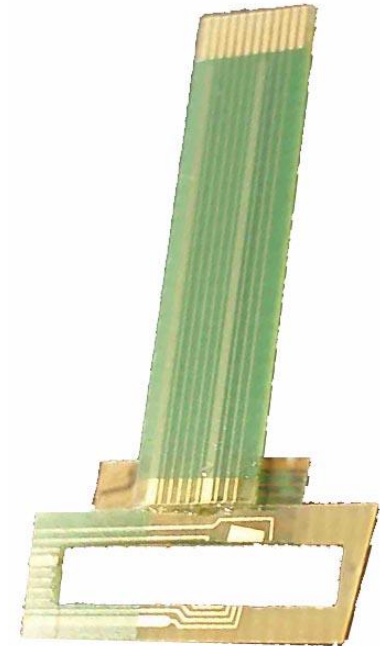
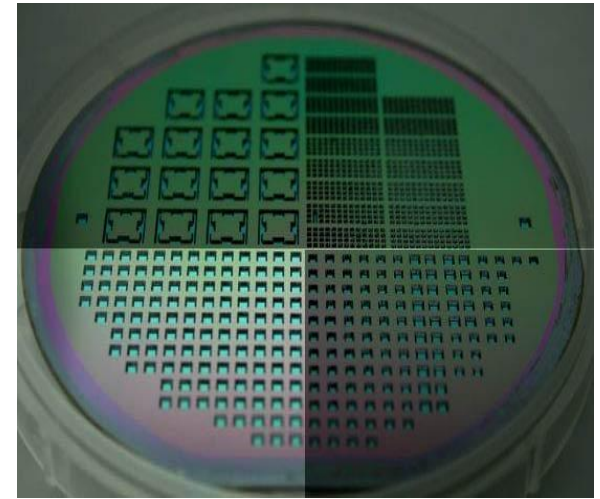
Improvements in version 2.0:

Better sound shielding

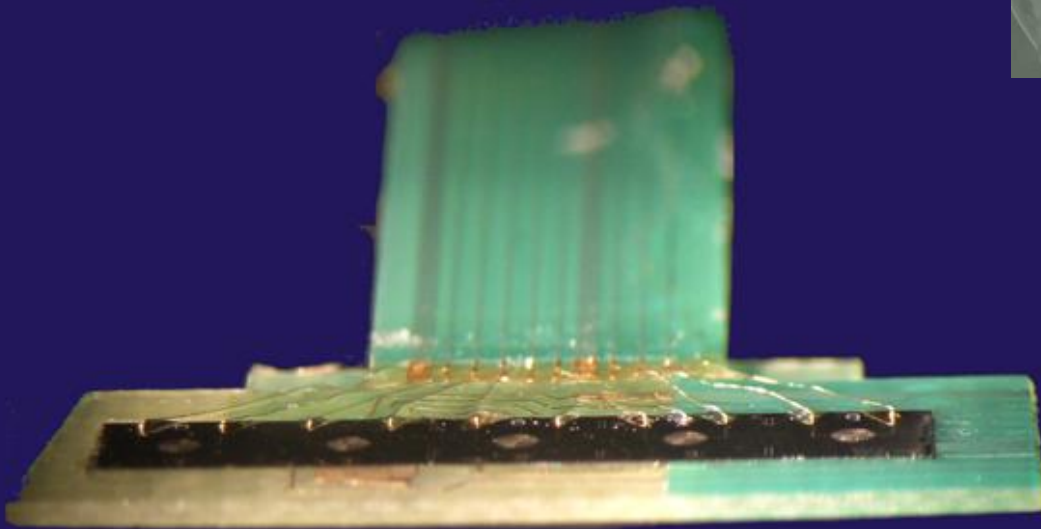
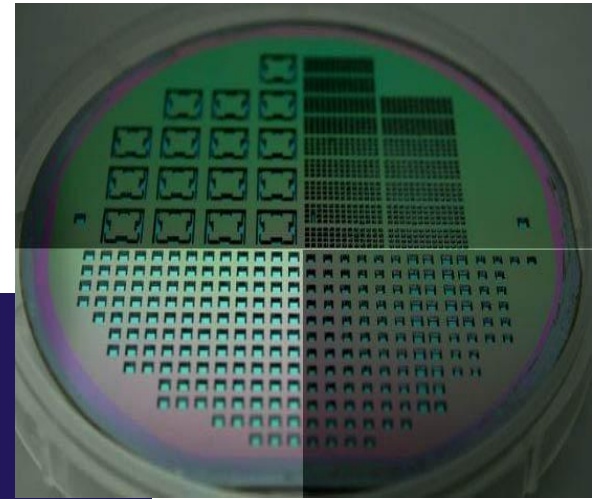
PCB all around

Easier bond wire configuration

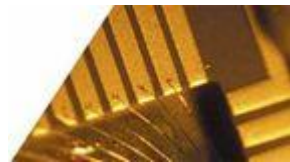
Turn the chip 90 deg



Ultrasonic microphone array prototype version 2.0

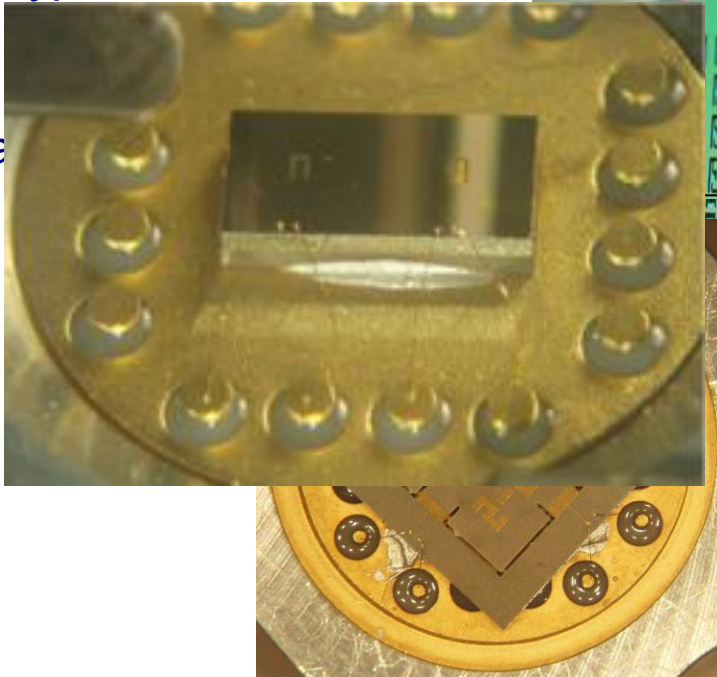


Fabricated in 10 samples and worked well



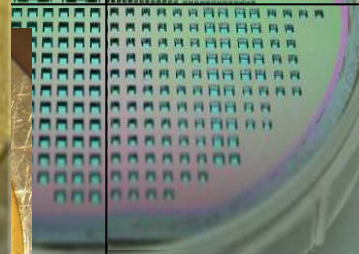
2 of the structures were put on hold after tests of version 1

Accelerometer
Prototype 1.0 was fabricated



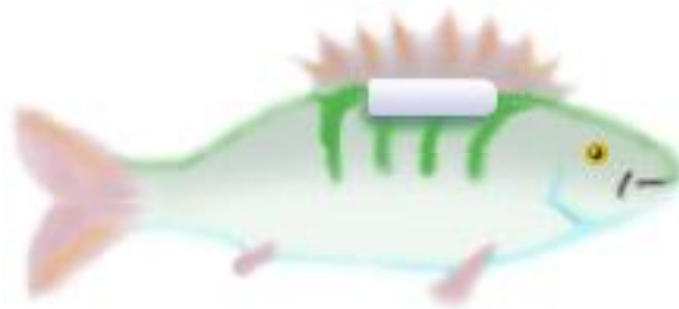
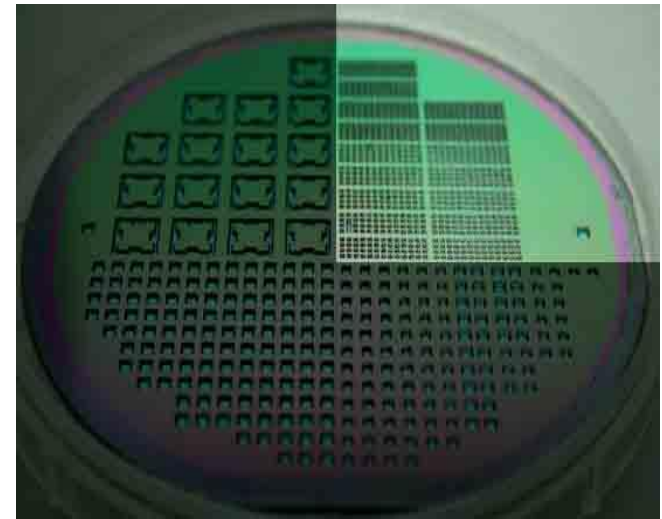
Due to a geometrical mismatch
the MEMS chip required a far too
small adjustable volume in the
suggested lid solution.

Acoustic gas sensors
When acoustic resonances
in the package itself gives that
version may be in LTCC

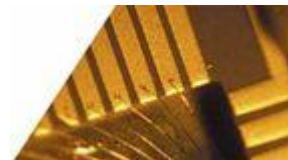


Fish tag transducers

Objective: Monitoring fish growth

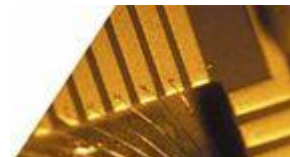
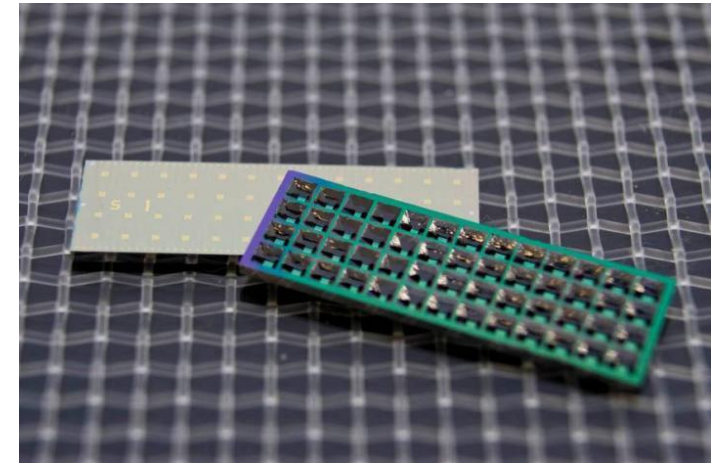
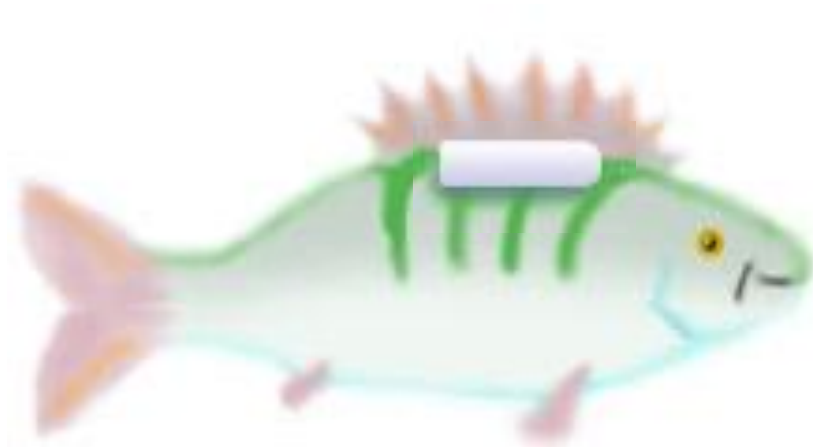
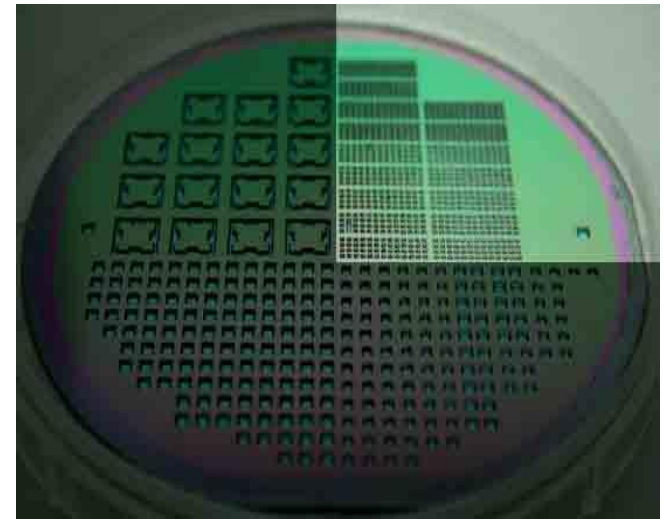


Tag

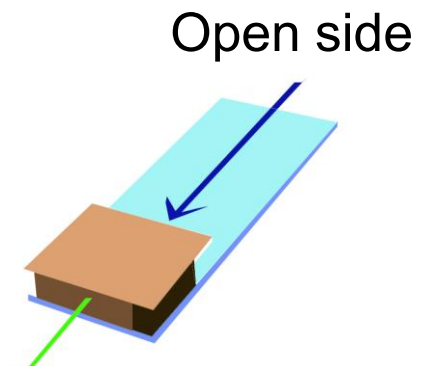
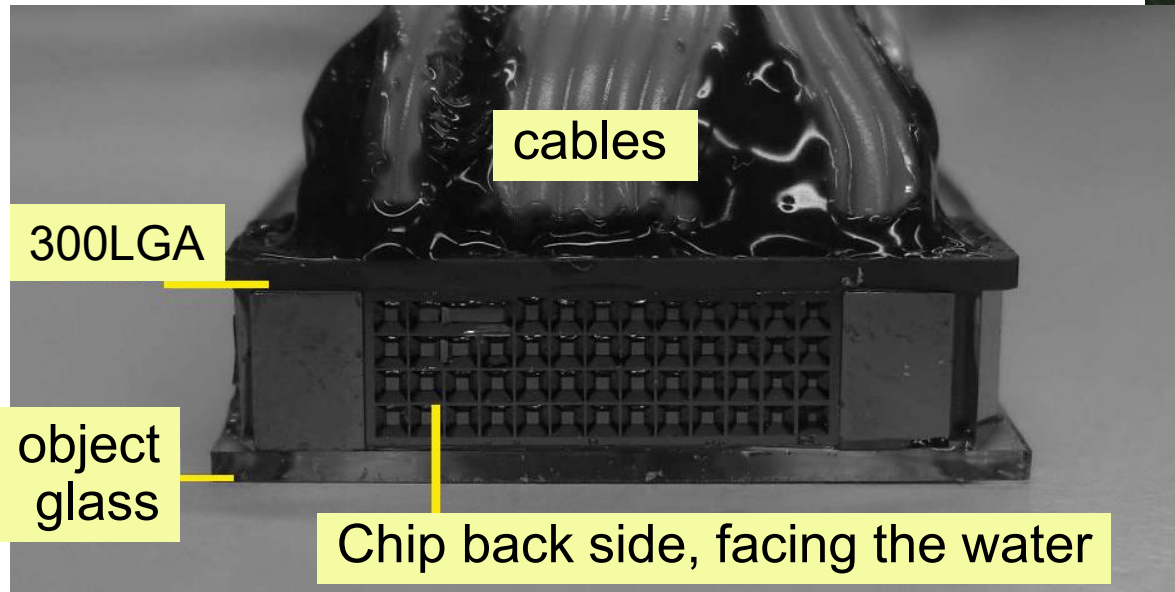
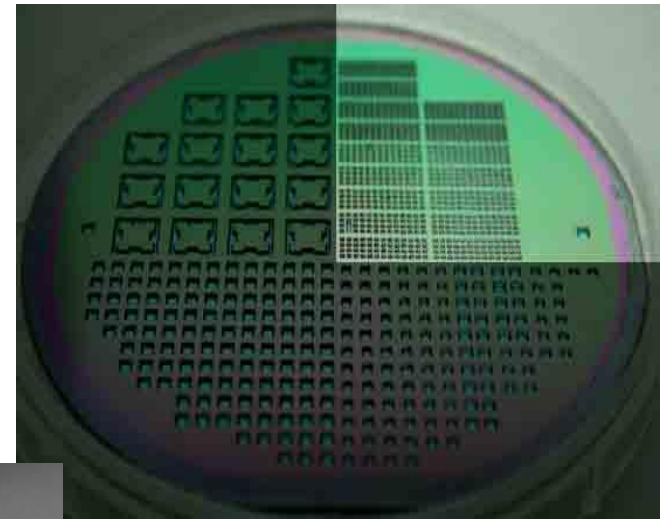


Fish tag transducers

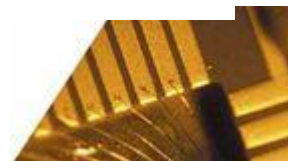
Objective: Monitoring fish growth



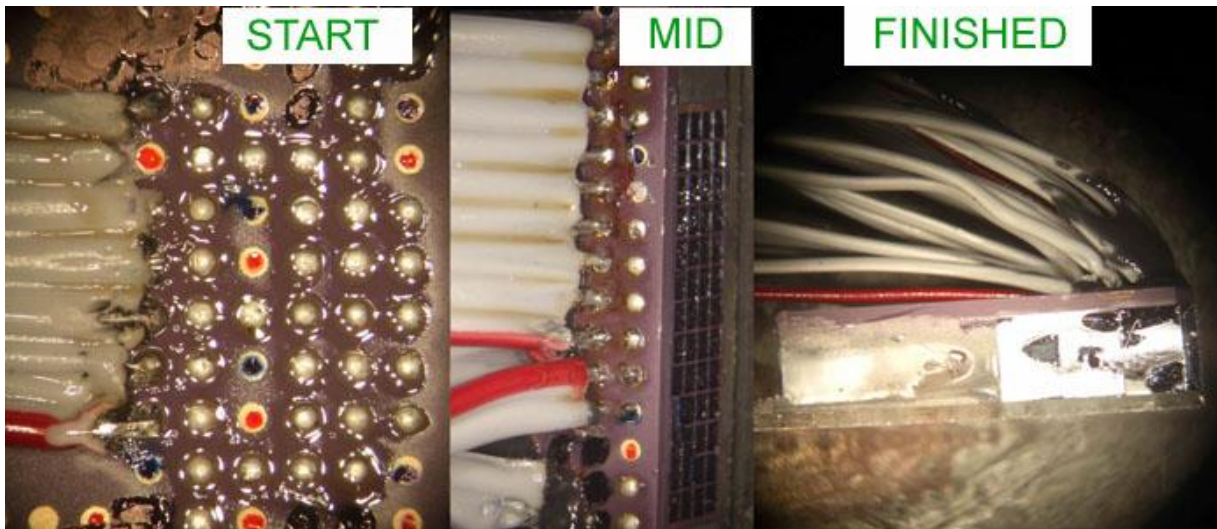
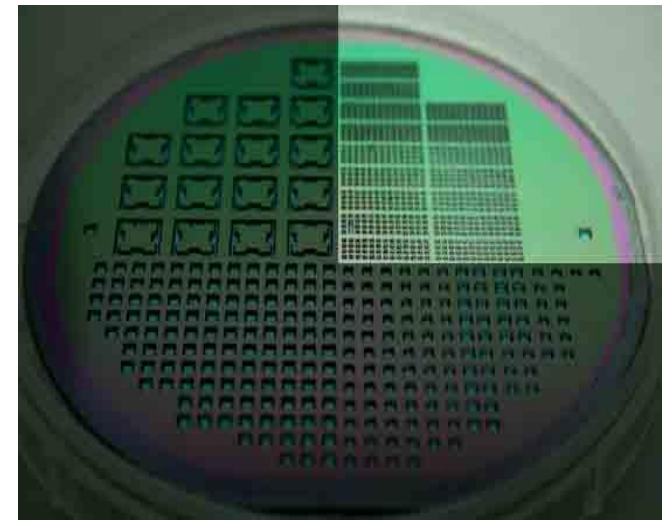
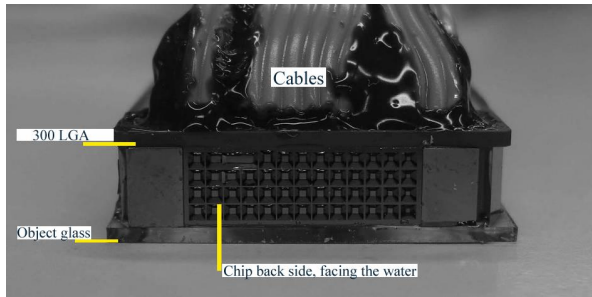
Fish tag transducers prototype version 1.0



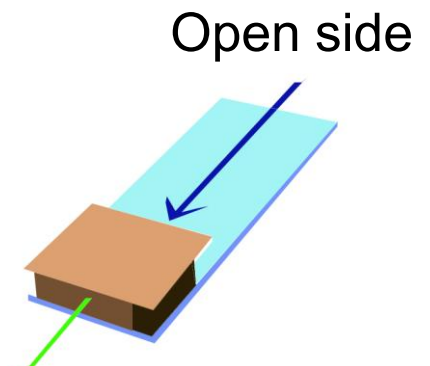
Chip back side



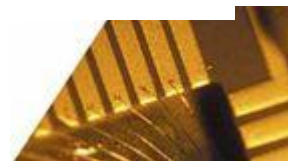
Fish tag transducers prototype version 1.0



The soldering process

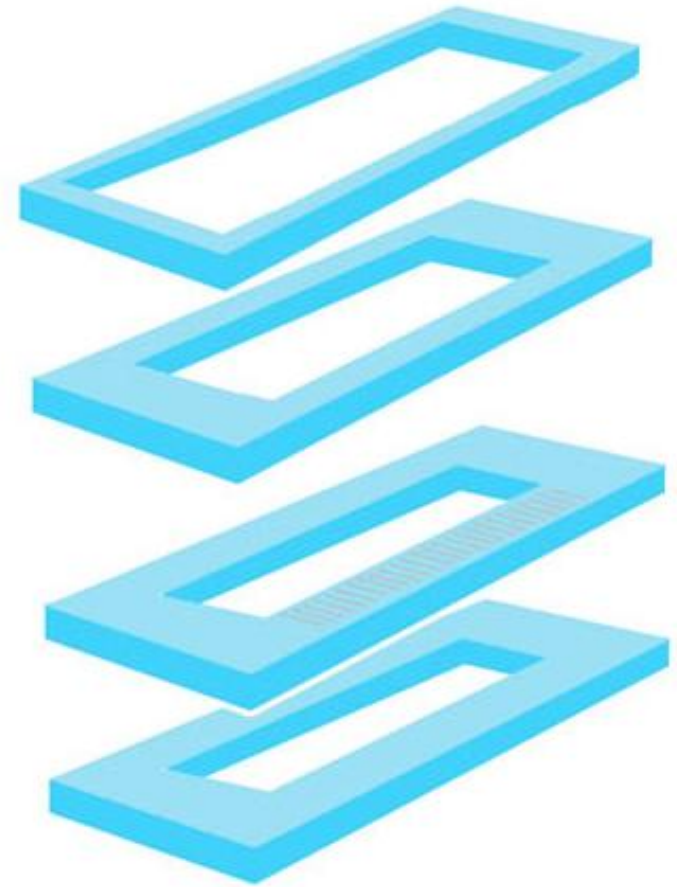
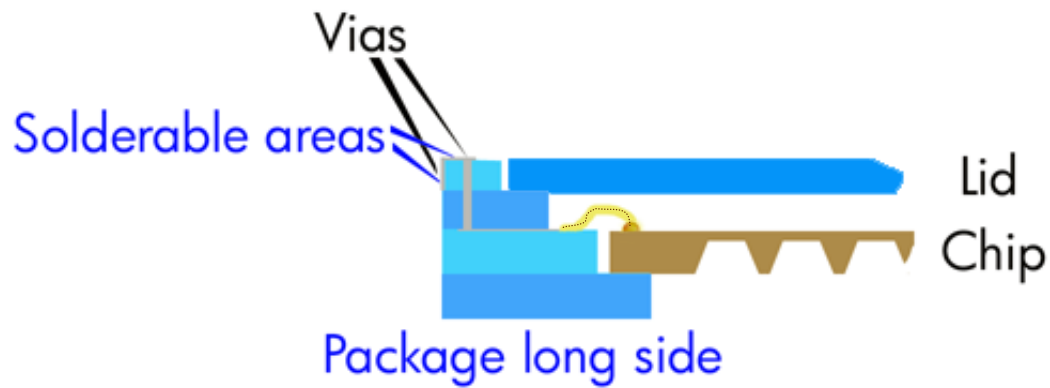


Chip back side



Fish tag prototype sketches ver. 2.0

Star-Oddi chip
package sketch

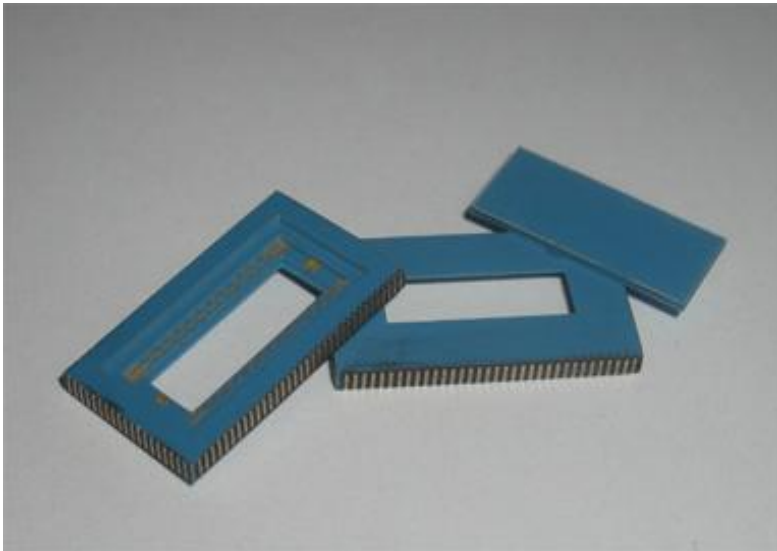


LTCC

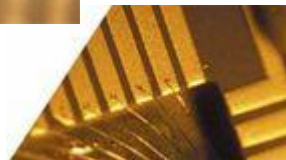
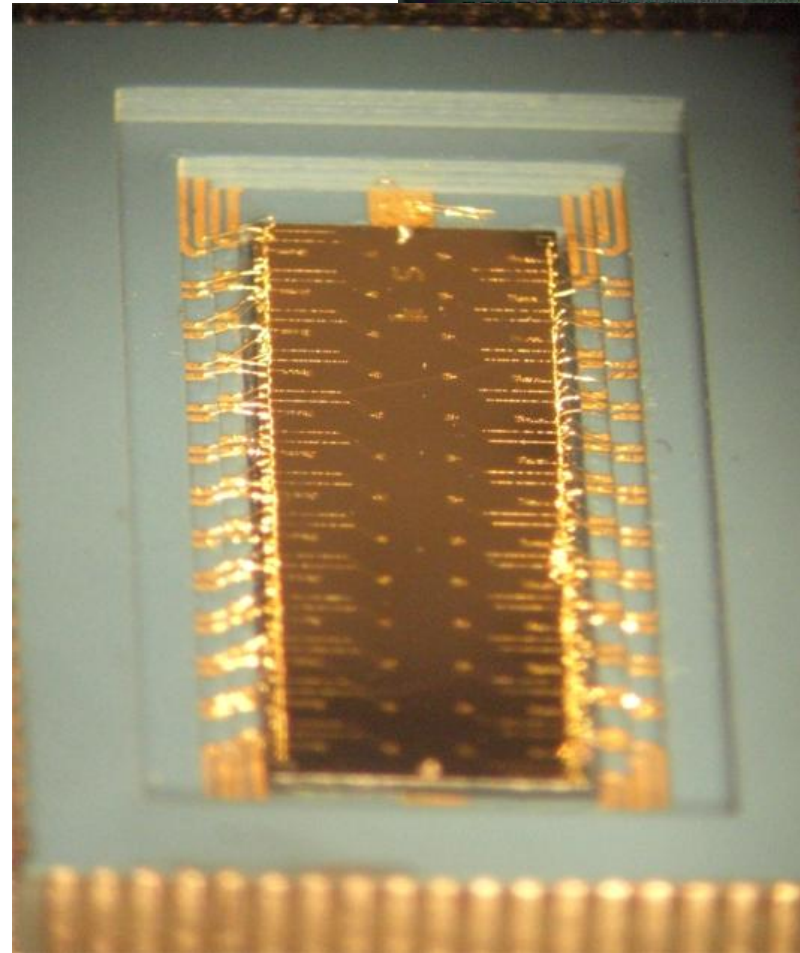
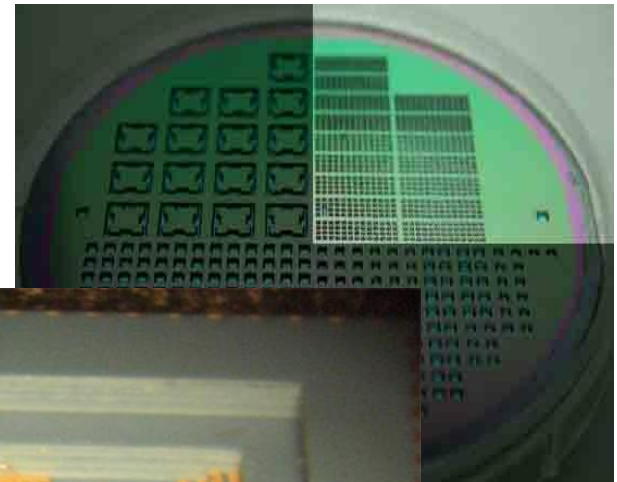


Fish tag

prototype version 2.0



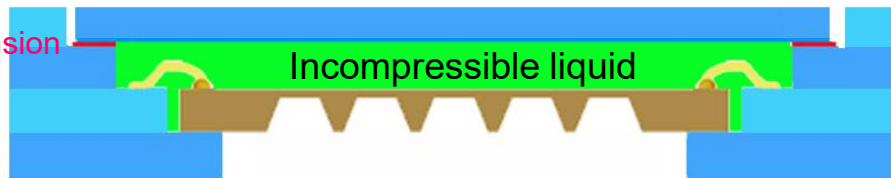
LTCC fabricated at Oulu University



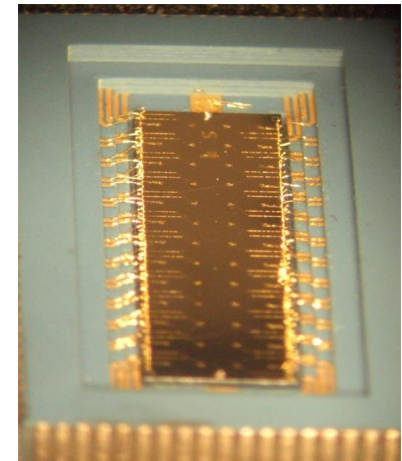
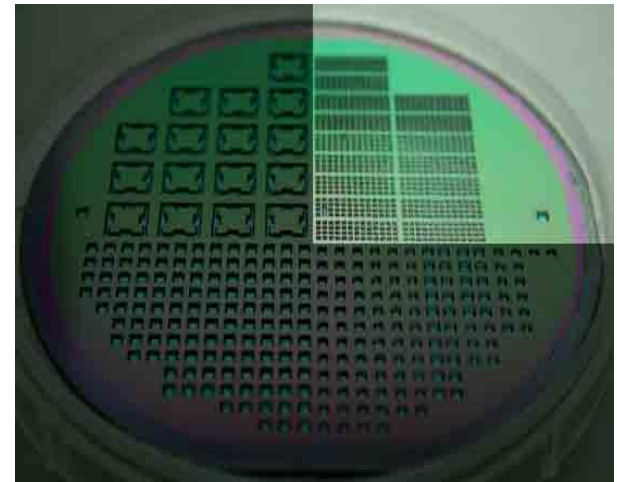
Fish tag

further prototype ideas

Pressure compensation



Ambient water pressure



Talk outline

Company presentation

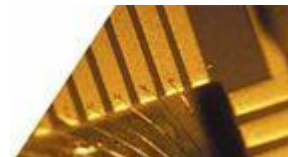
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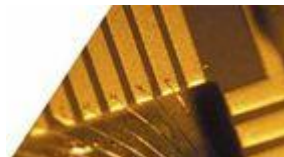
Conclusions

Questions



Advantages of the approach

- We could quickly adjust some designs after tests
- We were able to halt some designs without spending unnecessary time



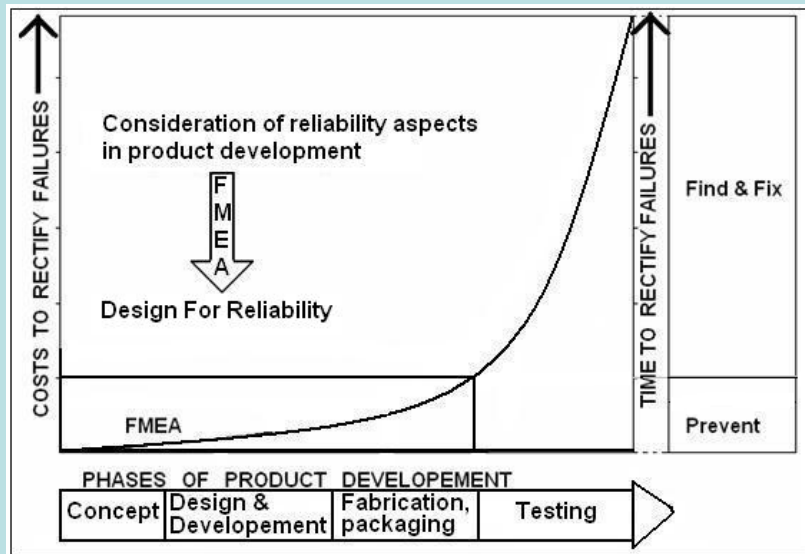
Conclusions

- The package may be splendid and the chip outstanding, but they have to fit each other in many aspects to do the job. Far too often, the fact that early system considerations saves later efforts is forgotten.
- In this NORD-pie project much time was spent early in order to avoid later patch work.



Another point of view

Advantages of FMEA approach

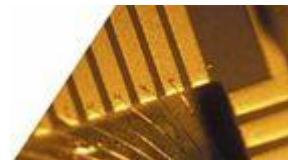


- To produce reliable MEMS, not only functionality but also reliability must be an integral aspect of development phases
- FMEA approach is pro-active rather than re-active
- Facilitates effective analysis of possible failures of multi physical background
- Successful application of FMEA avoids overlooking of problems in early stages of development and saves time and money.
- FMEA is a preventive technique and supports DfR (Design for Reliability) to ensure quality and reliability.
- Ensures ultimate design and reliable product.

In short, to reduce the total cost consider the whole system from the beginning

Simple idea.
-Yet so hard to implement

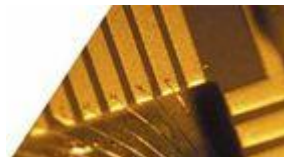
Figure with permission from an ESTC2008 short course recently



Curious about the name Mandalon?

- Imagine a three year old girl and a grapefruit
- She has no name for it...
- -It looks like a mandarin...
- ...but has the size of a melon!

Mandalon



Thank You for Your
attention!



This presentation is available on our home page www.mandalon.se

